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Stackable active sitting chair

Abstract

An active sitting chair is provided that encourages a sitter/user to sit actively by requiring balance adjustments and core engagement. The active sitting chair can be efficiently produced while being relatively sturdy and lightweight. The main body may be a monocoque shell that includes a rounded lower portion that contacts the surface the chair is on and promotes wobble of the chair when a sitter is seated on the chair. The chair may also include built-in handles for ease of lifting and a seat portion at the top of the chair that may include a cushion or similar component for increasing the stability and/or comfort of a sitter. The chair may also include a plurality of counterpart bosses and divots on opposing sides to facilitate stable stacking of the cylindrical active chairs.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to seating devices. In particular, the present invention provides a stackable ergonomic seat for promoting active sitting.

BACKGROUND

(2) Human bodies are built to move and generally require constant activity to remain supple and healthy. Unfortunately, modern life involves a good deal of sitting; indeed, many professions require many hours of simply sitting, which is an unnatural demand on the human bodies—so unnatural that children instinctively rebel against it.

(3) Sitting, and especially sitting still, aligns human bodies oddly, and denies joints the constant small adjustments that help to circulate the joint fluid which helps nourish the delicate cartilage lining of the joints. Additionally, sitting still denies core muscles the exercise involved in aligning and realigning our spines, exercise vital to keeping our core musculature strong and responsive. Moreover, extended and repetitive sitting has been linked to other health maladies. Indeed, the mismatch between our 21st-century built environment and our hunter-gatherer-optimized bodies has led to a variety of serious health problems: obesity, diabetes, heart disease, and even cancer, a litany of acquired diseases that can culminate in early death. These effects are not subtle—with a 50% increase in cancer risk among those who sat the most. Perhaps more importantly, simply reducing sitting could increase our lifespans by as much as two years.

(4) One potential solution to these health issues may be the use of active chairs, which are designed to cause users to sit actively and thus may allow for the recapture of the healthy blood chemistry of our hunter-gatherer forebears. These chairs allow for movement while sitting, and so allow users to sit all day as their modern lives require without suffering the harm brought on by sitting inertly. An example of one such an active chair is described in commonly owned U.S. Pat. No. 10,010,758. However, no single active chair is appropriate for all individuals or all situations, thus there is a need for active chairs provide for other types of motion and/or are appropriate for different users/situations.

SUMMARY OF THE DISCLOSURE

(5) An active sitting chair includes a monocoque shell having a bottom, a top, a front member, a back member, a first side member, a second side member opposite the first side member, a plurality of bosses on a first rim of the first side member, and a plurality of divots on a second rim of the second side member, wherein the front member connects and curves outwardly between the bottom and the top, wherein the back member connects and curves outwardly between the bottom and the top, wherein the first side member is connected to the bottom and the top, is connected to the front member and the back member, and curves inwardly between the bottom and the top, wherein the second side member is connected to the bottom and the top, is connected to the front member and the back member, and curves inwardly between the bottom and the top, and wherein each of the plurality of divots are configured to receive a respective one of each of the plurality of bosses when a pair of active sitting chairs are stacked with the first side member of one chair facing the second side member of another chair; a seat on the top; and a rounded lower portion attached to the bottom and arranged to rest on a supporting surface.

(6) A stackable active sitting chair includes a hollow, uninterrupted monocoque shell having a ring-shaped outer portion forming a continuous surface having a bottom portion, a top portion, a front portion, a rear portion, a first side edge having a first rim with a plurality of bosses, and a second side edge opposite the first side edge and having a second rim with plurality of divots. A first side portion is connected to the first side edge, wherein the first side portion curves inwardly from the first side edge. A second side portion is connected to the second side edge, wherein the second side portion curves inwardly from the second side edge. A rounded lower portion is attached to the bottom portion and arranged to rest on a supporting surface, wherein each of the plurality of divots are configured to receive a respective one of each of the plurality of bosses when a pair of active sitting chairs are stacked with the first side edge of one chair facing the second side edge of another chair.

(7) A method for stacking active sitting chairs includes placing a first chair on a surface and stacking a second chair on the first chair. The first chair and the second chair each include a

cylindrical shape and have a top portion, a bottom portion, a first side portion including a plurality of bosses, and a second side portion opposite the first side portion, wherein the second side portion includes a plurality of divots. The first chair and the second chair are aligned such that the first side portion of the first chair faces the surface and the first side portion of the second chair rests on the second side portion of the first chair such that each of the plurality of divots on the second side portion of the first chair engage with respective ones of each of the plurality of bosses on the first side portion of the second chair.

(8) A method for stacking active sitting chairs is provided that includes placing a first chair on a first side, wherein the first chair includes a monocoque shell having a bottom, a top, a front member, a back member, a second side opposite the first side, a plurality of divots on a first rim of the first side, and a plurality of bosses on a second rim of the second side, wherein the front member connects and curves outwardly between the bottom and the top, wherein the back member connects and curves outwardly between the bottom and the top, wherein the first side is connected to the bottom and the top, is connected to the front member and the back member, and curves inwardly between the bottom and the top, wherein the second side is connected to the bottom and the top, is connected to the front member and the back member, and curves inwardly between the bottom and the top, wherein the top includes a seat and the bottom includes a rounded portion configured to induce rocking when the first chair is in an upright position. A second chair is placed upon the first chair such that a first side of the second chair faces the second side of the first chair, wherein the second chair includes a monocoque shell having a bottom, a top, a front member, a back member, a second side opposite the first side, a plurality of divots on a first rim of the first side, and a plurality of bosses on a second rim of the second side, wherein the front member connects and curves outwardly between the bottom and the top, wherein the back member connects and curves outwardly between the bottom and the top, wherein the first side is connected to the bottom and the top, is connected to the front member and the back member, and curves inwardly between the bottom and the top, wherein the second side is connected to the bottom and the top, is connected to the front member and the back member, and curves inwardly between the bottom and the top, wherein the top includes a seat and the bottom includes a rounded portion configured to induce rocking when the first chair is in an upright position, such that the plurality of bosses on the second rim of the second side of the first chair engages with the plurality of divots on the first rim of the first side of the second chair. When stacked, the plurality of bosses on the second rim of the second side of the first chair and the plurality of divots on the first rim of the first side of the second chair are arranged around the respective second rim of the second side of the first chair and the first rim of the first side of the second chair such that the top of the first chair is aligned with the bottom of the second chair.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For the purpose of illustrating the invention, the drawings show aspects of one or more embodiments of the invention. However, it should be understood that the present invention is not limited to the precise arrangements and instrumentalities shown in the drawings, wherein:
- (2) FIG. 1 is a perspective view of an active sitting chair in accordance with an embodiment of the present invention;
- (3) FIG. 2 is a front view of the chair of FIG. 1;
- (4) FIG. 3 is a side view of the chair of FIG. 1;
- (5) FIG. 4 is a cross section view of the chair of FIG. 3;
- (6) FIG. 5 is a top view of the chair of FIG. 1;
- (7) FIG. 6 is a bottom of the chair of FIG. 1;

- (8) FIG. 7 is a perspective view of an active sitting chair in accordance with another embodiment of the present invention;
- (9) FIG. 8 is a front view of the chair of FIG. 7;
- (10) FIG. 9 is a side view of the chair of FIG. 7;
- (11) FIG. 10 is a cross section view of the chair of FIG. 9;
- (12) FIG. 11 is a top view of the chair of FIG. 7;
- (13) FIG. 12 is a bottom of the chair of FIG. 7;
- (14) FIG. 13 is a side perspective view of an active sitting chair with stacking bosses and divots in accordance with another embodiment of the present invention;
- (15) FIG. 14 is another side perspective view of the chair of FIG. 13;
- (16) FIG. 15 is a side view of the chair of FIG. 13;
- (17) FIG. 16 is a cross section view of the chair of FIG. 15;
- (18) FIGS. 17A-17B are perspective views of a plurality of chairs stacked in accordance with an embodiment of the present invention; and
- (19) FIGS. 18A-18B are cross section views of chairs stacked in accordance with an embodiment of the invention.

DESCRIPTION OF THE DISCLOSURE

(20) An active sitting chair is provided that encourages a sitter/user to sit actively by requiring balance adjustments and core engagement while sitting on the chair. The chair can be efficiently produced while being relatively sturdy and lightweight. The main body may be a monocoque shell that includes a rounded lower portion that contacts a supporting surface the chair is on and promotes wobbling of the chair when in use. The chair may also include built-in handles for ease of lifting/relocating and a seat portion at the top of the chair that may include a cushion or similar component for increasing the stability and/or comfort of a sitter.

(21) A ring or other similar structure may be included above the rounded lower portion to limit the extent of wobble to prevent the chair from tipping over. The monocoque structure provides structural integrity to support the weight of sitters while also allowing the chair to be hollow, which allows for the chair to be made efficiently and be relatively lightweight. Further, in a preferred embodiment, the front and rear portions connecting the rounded lower portion to the seat portion are convex and the side portions are concave. The concave shapes of the side portions provide additional structural integrity for supporting the weight of a sitter since the arch-like profile created is less likely to buckle. In another preferred embodiment, the profile of the chair when viewed from the front or back tapers such that the chair is narrower toward the top.

(22) Turning to the figures, an embodiment is shown in FIGS. 1-6 of an active sitting chair **100** that includes a hollow, single outer monocoque shell **104**, a seat portion **108**, a cushion **112** on seat portion **108**, a front portion **120A**, a back portion **120B**, a ring **124**, a rounded lower portion **128**, a first side portion **132A**, and a second side portion **132B**. A first indentation **116A** that is sized and configured to serve as a handle may be included on front portion **120A** and a second indentation **116B** that is sized and configured to serve as a handle may be included on back portion **120B**. Chair **100** is preferably hollow, with the design and shape of monocoque shell **104** providing the structural integrity necessary to support a sitter.

(23) Rounded lower portion **128** (as best seen in FIGS. 2, 3, and 6) rests on a supporting surface, such as a floor, when chair **100** is in use and is shaped and configured to promote wobbling motion when a sitter is sitting on chair **100**. Ring **124** (as can be seen in FIGS. 2-3) is positioned above rounded lower portion **128** in order to limit the range of tilt otherwise allowed by rounded lower portion **128** to avoid chair **100** tipping over. Each of front portion **120A** and back portion **120B** extends from rounded lower portion **128** up to seat portion **108** and are preferably convex, as can be seen in FIG. 3, in that they curve away from the center of chair **100** at the middle before curving back toward seat portion **108**.

(24) Preferably, and as shown in FIG. 4, first side portion **132A** and second side portion **132B** are

concave with respect to a vertical axis **102** of chair **100**, bending inward between connections at a lower part with rounded lower portion **128** and at an upper part with seat portion **108**. The concave profiles of first side portion **132A** and second side portion **132B** are less likely to buckle than straight sides and allow chair **100** to be hollow and shell **104** to be relatively thin while still supporting a sitter.

(25) The inward curving portions of the side members may be such that in a preferred embodiment a portion of each of the side members extends more than halfway to a vertical centerline **103** of the shell at least at the most extended portion as shown in FIG. 4.

(26) Shell **104** may be made of any suitable material, such as acrylonitrile butadiene styrene, and for use as an active chair may have a thickness of from about 0.32 cm to 1.27 cm, and preferably about 0.64 cm

(27) In another embodiment, as shown for example in FIGS. 7-12, an active sitting chair **200** includes a hollow, single outer monocoque shell **204** including a ring-shaped portion **206** having a seat area **208**, a front portion **230A**, and a back portion **230B**, a ring **224**, a rounded lower portion **228**, a first side portion **232A**, and a second side portion **232B**. Ring shaped portion **206** includes a rim or edge **210** on each side (e.g., **210A**, **210B**) to which first side portion **232A** and second side portion **232B** are respectively connected. Chair **200** is preferably hollow, with the design and shape of monocoque shell **204** providing the structural integrity necessary to support a sitter.

(28) Rounded lower portion **228** (as best seen in FIGS. 8, 9, and 12) rests on a supporting surface, such as a floor, when chair **200** is in use and is shaped and configured to promote wobbling motion when a sitter is sitting on chair **200**. Ring **224** (as can be seen in FIGS. 8-9) is positioned above rounded lower portion **228** in order to limit the range of tilt otherwise allowed by rounded lower portion **228** to avoid chair **200** tipping over. Each of front portion **230A** and back portion **230B** extends from rounded lower portion **228** up to seat portion **208** and are preferably convex, as can be seen in FIG. 9, in that they curve away from the center of chair **200** at the middle before curving back toward seat portion **208**.

(29) As can be seen in FIG. 10, first side portion **232A** and second side portion **232B** are preferably concave with respect to a vertical axis **202**, bending inward between connections at a lower part with rounded lower portion **228** and at an upper part with seat portion **208**. The concave profiles of first side portion **232A** and second side portion **232B** allow chair **200** to be hollow and shell **204** to be relatively thin while still supporting a sitter. Shell **204** may be made of any suitable material, such as acrylonitrile butadiene styrene, and for use as an active chair may have a thickness of from about 0.32 cm to 1.27 cm, and preferably about 0.64 cm.

(30) Chair **200** may preferably have a tapered profile from bottom to top, as can best be seen in FIGS. 8 and 11, in which when viewed from the front or back, the width of chair **200** is wider at the bottom than at the top.

(31) The inward curving portions of the side members may be such that in a preferred embodiment a portion of each of the side members extends more than halfway to a vertical centerline **203** of the shell at least at the most extended portion as shown in FIG. 10.

(32) In another embodiment, as shown for example in FIGS. 13-18, an active sitting chair **300** includes a hollow, single outer monocoque shell **304** including a ring-shaped portion **306** having a seat area **312** on a top portion, a lower ring **324** above a rounded lower portion **328**, a first side portion **332A**, and a second side portion **332B**. Ring shaped portion **306** includes a rim or edge **310** on each side (e.g., **310A**, **310B**) to which first side portion **332A** and second side portion **332B** are respectively connected. Chair **300** is preferably hollow, with the design and shape of monocoque shell **304** providing the structural integrity necessary to support a sitter.

(33) Rounded lower portion **328** rests on a supporting surface, such as a floor, when chair **300** is in use and is shaped and configured to promote wobbling motion when a sitter is sitting on chair **300**. Ring **324** is positioned above rounded lower portion **328** in order to limit the range of tilt otherwise allowed by rounded lower portion **328** to avoid chair **300** tipping over.

(34) Chair **300** may include the features of either chair **100** or chair **200** as described above, and may further include a plurality of bosses **350** (e.g., **350A-350C**) and a plurality of counterpart divots **352** (e.g., **352A-352C**). Bosses **350** are preferable spaced around rim or edge **310** (i.e., **310A**) with divots **352** spaced around opposing edge **310** (i.e., **310B**) such that when a first chair is stacked on a second chair with first side portion **332A** of one chair against second side portion **332B** of another, bosses **350** align with divots **352** to enable stable and efficient stacking of a plurality of chairs, as shown for example in FIGS. **17A-17B**. FIG. **18A** is a cross sectional view of three chairs stacked showing engagement of bosses **350** and divots **352** when stacked, with FIG. **18B** providing a detail view of such engagement. In this way, the generally round and wobbly chairs can be stably stacked on their sides.

(35) In a preferred embodiment, as shown in FIGS. **13-17B**, the divots and bosses on opposite sides of a given chair are configured to be staggered, i.e., positioned such that in order to align the divots and bosses of stacked chairs, the chairs are oriented oppositely with respect to tops and bottoms. In this way, the top of one chair will be facing the same direction as the bottom of the chair it is stacked together with (as can be seen in FIG. **17A**).

(36) Exemplary embodiments have been disclosed above and illustrated in the accompanying drawings. It will be understood by those skilled in the art that various changes, omissions, and additions may be made to that which is specifically disclosed herein without departing from the spirit and scope of the present invention.

Claims

1. An active sitting chair comprising: a monocoque shell having a bottom, a top, a front member, a back member, a first side member, a second side member opposite the first side member, a plurality of bosses on a first rim of the first side member, and a plurality of divots on a second rim of the second side member, wherein the front member connects and curves outwardly between the bottom and the top, wherein the back member connects and curves outwardly between the bottom and the top, wherein the first side member is connected to the bottom and the top, is connected to the front member and the back member, and curves inwardly between the bottom and the top, wherein the second side member is connected to the bottom and the top, is connected to the front member and the back member, and curves inwardly between the bottom and the top, and wherein each of the plurality of divots are configured to receive a respective one of each of the plurality of bosses when a pair of active sitting chairs are stacked with the first side member of one chair facing the second side member of another chair; a seat on the top, and a rounded lower portion attached to the bottom and arranged to rest on a supporting surface.
2. The active sitting chair of claim 1, wherein the rounded lower portion is shaped to induce a tilting motion of the chair about a vertical axis when a user is on the seat and the rounded lower portion is on the supporting surface.
3. The active sitting chair of claim 2, further including a ring positioned above the rounded lower portion and configured to limit a range of the tilting motion.
4. The active sitting chair of claim 2, further including a first indentation on the front member and a second indentation on the back member, wherein the first indentation and second indentation are shaped to accommodate a grasp of the user.
5. The active sitting chair of claim 2, wherein the shell is of unitary construction and is hollow.
6. The active sitting chair of claim 5, wherein the shell is less than 1 cm thick.
7. The active sitting chair of claim 5, wherein the front member includes a front member bottom width and a front member top width, wherein the front member bottom width is wider than the front member top width, wherein the back member includes a back member bottom width and a back member top width, and wherein the back member bottom width is wider than the back member top width.

8. The active sitting chair of claim 5, wherein the first side member extends more than half way to a vertical centerline of the shell.
 9. The active sitting chair of claim 5, wherein the shell does not include any holes passing from the first side member to the second side member.
 10. A stackable active sitting chair comprising: a hollow, uninterrupted monocoque shell including: a ring-shaped outer portion forming a continuous surface having a bottom portion, a top portion, a front portion, a rear portion, a first side edge having a first rim with a plurality of bosses, and a second side edge opposite the first side edge and having a second rim with plurality of divots; a first side portion connected to the first side edge, wherein the first side portion curves inwardly from the first side edge; and a second side portion connected to the second side edge, wherein the second side portion curves inwardly from the second side edge; and a rounded lower portion attached to the bottom portion and arranged to rest on a supporting surface, wherein each of the plurality of divots are configured to receive a respective one of each of the plurality of bosses when a pair of active sitting chairs are stacked with the first side edge of one chair facing the second side edge of another chair.
 11. The stackable active sitting chair of claim 10, wherein the ring-shaped outer portion has a width and wherein the width is narrower at the top portion than at the bottom portion.
 12. The stackable active sitting chair of claim 10, wherein the first side portion extends more than half way to a vertical centerline of the ring-shaped outer portion and wherein the second side portion extends more than half way to the vertical centerline of the ring-shaped outer portion.
 13. The stackable active sitting chair of claim 10, further including a seat on the top portion, wherein the rounded lower portion is shaped to induce a tilting motion of the shell about a vertical axis when a user is on the seat and the rounded lower portion is on the supporting surface.
 14. The stackable active sitting chair of claim 13, further including a ring positioned above the rounded lower portion and configured to limit a range of the tilting motion.
 15. The stackable active sitting chair of claim 14, further including a first indentation on the front portion and a second indentation on the rear portion, wherein the first indentation and second indentation are shaped to accommodate a grasp of the user.
 16. The stackable active sitting chair of claim 14, wherein the shell is less than 1 cm thick at all points.
 17. The stackable active sitting chair of claim 14, wherein the shell does not include any holes larger than 1 cm in diameter.
 18. A method for stacking active sitting chairs comprising: placing a first chair on a surface; stacking a second chair on the first chair, wherein the first chair and the second chair each include a cylindrical shape and have a top portion, a bottom portion, a first side portion including a plurality of bosses, and a second side portion opposite the first side portion, wherein the second side portion includes a plurality of divots; and aligning the first chair and the second chair such that the first side portion of the first chair faces the surface and the first side portion of the second chair rests on the second side portion of the first chair such that each of the plurality of divots on the second side portion of the first chair engage with respective ones of each of the plurality of bosses on the first side portion of the second chair, wherein the first chair and the second chair each include a hollow, uninterrupted monocoque shell, wherein the first side portion curves inwardly, wherein the second side portion curves inwardly, and wherein the first chair and the second chair each include a rounded lower portion attached to the bottom portion configured to induce rocking when in an upright position.
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